GUARDIAN-AI

Smart ADAS and Real-Time Driver Safety Copilot

Hack2Skill Data & AI Challenge Submission | Built by Jatin

The Problem: Cognitive Distractions & Road Hazards

1.3 Million Annual Road Deaths: Global statistics show that the vast majority of severe vehicular accidents are caused by human error, primarily micro-sleeps or distractions.

- Silent Fatigue (Micro-sleeps): Drivers fall asleep for small intervals, leading to immediate lane drift and high-speed crashes.

- Visual Distractions (Phone Usage): Looking away to interact with devices blocks response times for obstacle
avoidance.

- Missing Immediate Closed-Loop Warnings: Existing dashboard cameras only record feeds, lacking edge-computer vision models that calculate risk scores and warning signals locally in real-time.

GUARDIAN-AI | SAFETY IN TRANSIT Slide 2 of 10

Executive Summary: The GuardianAI Vision

GuardianAI bridges the critical gap between video capturing and physical accident prevention. By using lightweight edge computer vision in the browser and asynchronous multi-agent models in the backend, it acts as a proactive safety co-pilot.

- Attentiveness Indexing: Continuously scores driver focus from 0 to 100 based on blink frequency, head angles, and
yaw deviation.
- Asynchronous Telemetry: Combines driver posture with live steering and braking loops to calculate stability.
- Collision Forecasting: Projects trajectory risk 3-5 seconds before impact, giving drivers enough time to take corrective action.

GUARDIAN-AI | SAFETY IN TRANSIT Slide 3 of 10

System Architecture: 4 Collaborating AI Agents

Our system implements an event-driven micro-agent architecture running asynchronously on a central event bus:

[Driver Behavior Agent]
Monitors face yaw, pitch, blink rate (EAR), and
smartphone phone call postures.

[Vehicle Dynamics Agent]
Tracks speed, steering angle deviations, and road lane
alignment values.

[Prediction Agent]
Fuses driver and vehicle features to forecast accident
probability and type.

[Intervention Agent]
Escalates warnings (alarms, automatic deceleration instructions) to prevent crashes.

GUARDIAN-AI | SAFETY IN TRANSIT Slide 4 of 10

Client-Side Edge AI: MediaPipe Holistic CV

Rather than streaming heavy video feeds to the cloud, GuardianAI runs all computer vision models locally inside the driver's browser interface:

- Yaw & Pitch angle estimation: Uses scale-invariant nose-tip to eye ratios to catch left/right turns and up/down nodding.
- Dual-Cheek proximity detection: Checks wrist and finger landmarks to detect phone usage at either ear.
- Zero Server GPU Cost: Processing frames locally reduces backend workload and solves bandwidth constraints.

GUARDIAN-AI | SAFETY IN TRANSIT Slide 5 of 10

Backend Intelligence: PyTorch Neural Networks

All backend models are fully trained in PyTorch on generated sequence scenarios and saved as checkpoints:

Model Name Architecture Type File Checkpoint
Driver Monitor MobileNetV2 CNN Classifier driver_monitor.pt
Vehicle Dynamics Bi-directional GRU Sequence vehicle_behavior.pt
Sensor Fusion Transformer Cross-Attention sensor_fusion.pt
Accident Predictor LSTM Sequence Forecaster accident_predictor.pt

GUARDIAN-AI | SAFETY IN TRANSIT Slide 6 of 10

Safety Interventions Escalation Ladder

The Intervention Agent maps real-time risk scores directly to safety controls:

Risk Score Severity System Intervention Action
0 - 25% LOW Normal operations, background monitoring active.
25 - 50% MEDIUM Caution advisory tips played on screen, voice guidance starts.
50 - 75% HIGH Warning wailing sirens sound, lane alignment advisory active.
75 - 100% CRITICAL High-pitch wailing siren, automatic deceleration instruction sent.

GUARDIAN-AI | SAFETY IN TRANSIT Slide 7 of 10

Interactive Cockpit Controls & DOM Optimization

- Telemetry Override sliders: Control speed, steering angle, and lane alignment variables directly in the dashboard UI to simulate emergency risk conditions.
- Direct DOM Node Updates: Excludes high-frequency eye and landmark values from the React state cycle. Coordinates are updated directly via element IDs, preventing interface lag.
- Throttled Sockets: Transfers webcam parameters to the backend at 150ms intervals, avoiding websocket buffers overloading.

GUARDIAN-AI | SAFETY IN TRANSIT Slide 8 of 10

System Verification & Test Metrics

- 10/10 pytest Suite: Verifies asynchronous event queues, database persistence logs, API routes, and network models successfully.
- Inference Timing: Neural network forward passes execute in under 10ms on CPU, leaving ample margin within the real-time requirements.
- Robust Fallback: If model checkpoints are missing, the agents fallback automatically to rule-based simulations to prevent system crashes.

GUARDIAN-AI | SAFETY IN TRANSIT Slide 9 of 10

Summary & Future Vision

GuardianAI represents a scalable, reliable driver assistance system designed for mass adoption:

- Mass Adoption: Low CPU footprint allows it to run on entry-level dashboards and standard smartphones.
- Cloud Deployments: Fully configured for Vercel deployment with dynamic WebSocket proxying.
- Future V2X Integration: Connects to surrounding cars over 5G to negotiate lane changes during emergency brake

maneuvers.

GUARDIAN-AI | SAFETY IN TRANSIT Slide 10 of 10